Internet of Things (IoT) with Applications

Undergraduate Engineering Course

Module 4

IoT Security and Data Analytics

Securing the Stack, Common Threats, and Turning Data into Insight

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Unit IV: IoT Security and Data Analytics

Concepts

Topics Covered:

Security Requirements: Authentication, Authorization, Encryption
Security by Layer: Device, Network, Cloud · Threats: Spoofing, DoS, Malware
IoT Data Analytics: Preprocessing, Visualization, Predictive Analytics
Blockchain + IoT: Decentralization, Smart Contracts, Identity Management

1 Introduction

Modules 1 through 3 built up a complete picture of how an IoT deployment senses, transmits, and processes data, from a soil-moisture sensor through to a cloud dashboard. Every one of those components – the device itself, the wireless link, the gateway, and the cloud platform – is also a potential point of attack, and the very characteristics that make IoT powerful (scale, heterogeneity, and often unattended operation) are precisely what make it difficult to secure. A single compromised sensor node is rarely catastrophic on its own, but an IoT deployment's scale means a single vulnerability, once found, can often be replicated across thousands of identical devices at once.

This module addresses two related concerns. Sections 2 through 4 cover **IoT security**: the requirements a secure deployment must satisfy, how those requirements map onto each layer of the architecture from Module 1, and the specific threats a deployment must defend against. Sections 5 and 6 then turn from protecting data to extracting value from it, covering **IoT data analytics** – the pipeline that turns raw, noisy sensor readings into decisions – and **blockchain**, an emerging approach to guaranteeing data integrity and device identity without relying on a single trusted central authority.

2 IoT Security Requirements

Before examining specific defences, it is useful to state precisely what a secure IoT system is trying to achieve. Four requirements recur across nearly every treatment of IoT security:

- **Confidentiality:** Sensor readings and control commands should be readable only by parties authorised to see them – a patient's heart-rate data, for instance, should not be visible to anyone other than their care team and the patient themselves.
- **Integrity:** Data must not be alterable, in transit or at rest, without detection – a tampered temperature reading from a cold-chain shipment sensor could hide a genuine spoilage event.
- **Availability:** The system must remain operational and responsive under normal and adversarial conditions alike – a security system that can be trivially knocked offline (Section 4.4) provides little real protection.
- **Authenticity:** A message must be verifiably attributable to the specific device (or user) that sent it – without this, an attacker could impersonate a legitimate sensor and inject false readings.

2.1 Authentication and Authorization

Authentication answers the question "who – or what device – is this?", while **authorization** answers the separate question "what is this identified party allowed to do?". In an IoT context, authentication is typically handled by issuing each device a unique cryptographic identity (a certificate or pre-shared key) at manufacture or provisioning time, verified whenever the device connects to the network or cloud platform. Authorization then determines, for example, whether an authenticated sensor is permitted only to *publish* readings, or whether an authenticated maintenance technician's app is additionally permitted to *push* configuration changes – the two concepts are independent, and a system can authenticate a device correctly while still enforcing that it is authorized to perform only a narrow set of actions.

2.2 Encryption and Secure Communication

Encryption protects confidentiality by transforming data so that it is unreadable without a corresponding decryption key, and is normally applied both to **data in transit** (as it crosses the network links of Module 2) and to **data at rest** (as it sits in cloud storage, Module 3). **Secure communication** protocols – such as TLS/DTLS running underneath MQTT or CoAP – combine encryption with the authentication and integrity checks above into a single, standard transport, so that individual applications do not need to re-implement cryptography themselves.

Worked Example

Distinguishing authentication from authorization. A smart-lock system verifies a delivery courier's app using a one-time cryptographic token issued for a specific delivery window, and grants that app permission only to unlock the front door, not the interior office doors, and only within that time window. Identify which part of this scenario is authentication and which is authorization.

Solution. Verifying the courier's app using the cryptographic token is authentication – confirming which identity is making the request. Restricting that identity to unlocking only the front door, and only within the delivery window, is authorization – controlling what the authenticated identity is permitted to do.

3 Security Across the IoT Stack

Because an IoT deployment spans devices, networks, and cloud infrastructure (Modules 1 and 3), security cannot be addressed at a single point – it must be layered, matching the Perception/Network/Processing structure of the reference architecture introduced in Module 1.

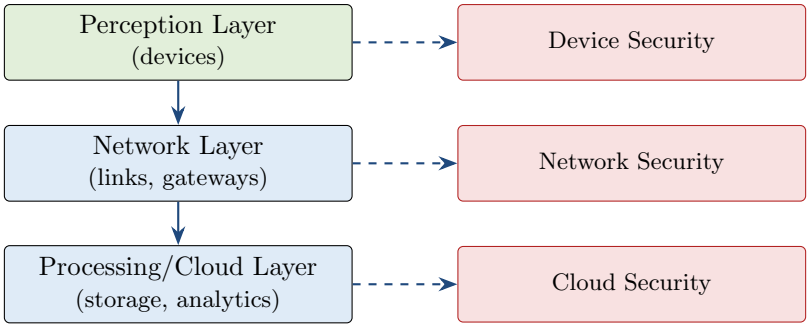


Figure 1: Security requirements mapped onto the three-layer IoT reference architecture from Module 1: each layer has a corresponding security concern that must be addressed independently, since a defence at one layer does not protect the others.

3.1 Device Security

Individual devices are frequently the weakest point in an IoT deployment: many are physically accessible to an attacker (unlike a data centre server), are manufactured to a tight cost budget that limits on-board cryptographic capability, and are often deployed with **default credentials** left unchanged. Device security therefore includes secure boot (verifying that only authorised firmware runs), hardware-backed key storage where feasible, and processes for applying security patches to already-deployed devices – itself complicated by the scale and often-limited connectivity of IoT

fleets.

3.2 Network Security

Network security protects data as it moves across the short- and long-range links of Module 2 – through encryption (Section 2.3), and through network-level defences such as firewalls at the gateway and segmentation of IoT traffic away from an organisation's other network traffic, so that a compromised sensor cannot be used as a stepping-stone to attack unrelated systems.

3.3 Cloud Security

Cloud security protects the storage and analytics infrastructure of Module 3: access control over who (and which services) can query or modify stored device data, encryption of data at rest, and monitoring for anomalous access patterns that might indicate a compromised account or API key – notably, the same anomaly-detection techniques used on sensor data (Module 3, Section 5) are often applied to access logs themselves.

4 IoT Threats

Understanding the security requirements of Section 2 and the layered defences of Section 3 is easier with a concrete sense of what they are defending *against*. This section surveys the threat types most specific to, or most amplified by, the IoT setting.

Threat	**Description**	**Primarily Targets**
Device attacks	Physical tampering or exploitation of a device's firmware/hardware to extract keys or alter behaviour.	Perception Layer / Device Security
Eavesdropping	Passively intercepting unencrypted or weakly encrypted traffic to read sensitive data.	Network Layer / Confidentiality
Spoofing	Impersonating a legitimate device or user to inject false data or commands.	Network/Cloud Layer / Authenticity
Denial-of-Service (DoS)	Overwhelming a device, gateway, or cloud service with traffic or requests so it cannot serve legitimate users.	Any layer / Availability
Malware	Malicious software installed on a device or gateway, often to recruit it into a larger botnet.	Perception Layer / Device Security
Data leakage	Unauthorised exposure of stored or transmitted data, often via misconfigured cloud storage or weak access control.	Cloud Layer / Confidentiality

Table 1: Common IoT threat categories, mapped to the layer and requirement they primarily affect.

Threat Spotlight

Botnets built from compromised IoT devices. A recurring real-world pattern combines several of the threats above: an attacker uses default or weak device credentials (a device-security failure) to install malware on large numbers of IoT devices, then coordinates those devices into a botnet that launches a denial-of-service attack against an unrelated third-party target. This illustrates why device security cannot be treated as a purely local concern – a single insecure device model, deployed at scale, becomes a shared liability for the wider Internet, not just for its own owner.

Worked Example

Classifying an incident. An attacker intercepts unencrypted MQTT traffic between a factory's temperature sensors and its gateway, and separately begins publishing fabricated temperature readings under a legitimate sensor's device ID to mask a real overheating event. Identify the two threat types involved, and state which security requirement from Section 2 each one violates.

Solution. Intercepting the unencrypted traffic is eavesdropping, violating confidentiality. Publishing fabricated readings under a legitimate device's identity is spoofing, violating authenticity (and, since the readings are false, also integrity of the data the system relies on).

5 IoT Data Analytics

Raw sensor data, even once safely collected, is rarely useful in its rawest form: it is noisy, arrives at irregular intervals, and vastly outweighs what a person can review manually. This section extends the "cloud analytics" step of Module 3 into a fuller analytics pipeline.



Figure 2: The IoT data analytics pipeline. Preprocessing and time-series structuring feed both human-facing visualization and machine-learning-based predictive analytics.

5.1 Data Preprocessing

Data preprocessing cleans and prepares raw sensor readings before any analysis: removing or flagging obviously invalid readings (a temperature sensor briefly reporting $-999\,^{\circ}\mathrm{C}$ on a fault), handling missing values from dropped transmissions, and resampling irregularly-timed readings onto a consistent time grid. Analytics performed on unpreprocessed data risks drawing conclusions from sensor faults and transmission gaps rather than genuine physical phenomena.

5.2 Time-Series Data

As introduced in Module 3, IoT data is predominantly **time-series**: a sequence of values indexed by timestamp. Time-series-specific techniques – identifying trend, seasonality (e.g., a daily temperature cycle), and irregular residual variation – underpin both visualization and predictive analytics, since a naive analysis that ignores known seasonal patterns can mistake ordinary daily variation for an anomaly.

5.3 Visualization

Visualization – line charts, heatmaps, geographic overlays – turns processed time-series data into a form a human can interpret quickly, and is the analytical counterpart to the dashboards introduced in Module 3: where a dashboard shows current status, good visualization of historical data reveals trends and patterns that raw numbers or a single current reading would not.

5.4 Predictive Analytics and AI/ML-Based Analytics

Predictive analytics uses historical data to forecast future values or events – extending the anomaly detection and predictive maintenance techniques introduced in Module 3 with a fuller data-engineering pipeline behind them. In practice, this stage depends entirely on the quality of the preprocessing and time-series structuring that precede it: an AI/ML model trained on noisy, unaligned data will inherit and often amplify that noise in its predictions, which is why Sections 5.1 and 5.2 are treated as prerequisites to Section 5.4 rather than optional steps.

Worked Example

Why preprocessing matters. A predictive-maintenance model is trained directly on raw vibration sensor data that includes a two-week period of missing readings (due to a gateway outage) left as zero-valued entries, rather than being flagged and excluded or interpolated.

Explain what is likely to go wrong, and how Section 5.1 would prevent it.
Solution. Leaving missing readings as zero-valued rather than flagging them introduces a large block of artificial "zero vibration" data that does not reflect the machine's actual state during the outage; a model trained on this data may learn to associate this artificial pattern with normal operation, degrading its ability to detect genuine low-vibration anomalies. Data preprocessing that explicitly flags and either excludes or properly interpolates the missing period would prevent this artificial data from being treated as genuine.

6 Blockchain + IoT

A recurring weakness in the security model of Sections 2–4 is that it typically relies on a *central* authority – a certificate authority, a cloud provider's access-control system – as the root of trust. **Blockchain** technology offers an alternative: a distributed, cryptographically-linked ledger maintained collectively across many participants, with no single party able to unilaterally alter recorded history.

6.1 Decentralized IoT

In a **decentralized IoT** architecture, device identities, transaction records, or data hashes are recorded on a blockchain rather than in a single company's central database, removing the single point of failure and single point of trust that a conventional cloud-centric architecture depends on. This comes at the cost of the additional computational and storage overhead blockchain participation requires – typically too heavy for a constrained sensor node itself, so blockchain integration is usually implemented at the gateway or cloud tier rather than on the device.

6.2 Smart Contracts

A **smart contract** is self-executing code stored on a blockchain that runs automatically when predefined conditions are met, without requiring a trusted intermediary. In an IoT context, this could mean an agricultural supply-chain contract that automatically releases payment once a shipment's temperature sensor confirms it remained within a safe range for the entire journey – the sensor data itself becomes the trigger for an automated, tamper-resistant business process.

6.3 Data Integrity

Because blockchain records are cryptographically linked and distributed, altering a previously recorded entry would require simultaneously altering every subsequent linked record across a majority of participants – making tampering computationally impractical after the fact. Recording a hash of sensor data on a blockchain, even if the raw data itself is stored elsewhere, provides a tamper-evident record: any later modification to the stored data would no longer match its recorded hash.

6.4 IoT Identity Management

Blockchain-based **identity management** allows each device to hold a self-sovereign, cryptographically verifiable identity that does not depend on a single issuing authority remaining trustworthy and available – addressing, at least partially, the device-authentication requirement from Section 2.2 without the single point of failure a centralised certificate authority represents.

IoT in Practice: Connecting Back to the Course

This module closes the loop on the "make it trustworthy and useful" half of the course. Sections 2–4 secure the exact devices, links, and cloud platform built up in Modules 1–3, mapping the authentication, encryption, and layered defences directly onto the Perception/Network/Processing architecture first introduced in Module 1. Sections 5 and 6 then extend Module 3's cloud analytics and Edge AI into a fuller data pipeline and an alternative, decentralised trust model. Unit V now assembles every module into complete, secured, analytics-driven applications: a smart-agriculture deployment protected by device authentication and analysed through the pipeline of Section 5; a healthcare system whose patient data confidentiality depends directly on the encryption and access-control principles of Sections 2 and 3; and supply-chain applications that draw specifically on the blockchain concepts of Section 6.

7 Solved Problems

Worked Example

Problem 1. A hospital's remote patient-monitoring system transmits ECG data encrypted end-to-end, but every device on the network shares a single hard-coded password set at the factory. Which security requirement from Section 2 is well addressed here, and which is dangerously weak? Justify your answer.

Solution. Confidentiality is well addressed by the end-to-end encryption of the ECG data in transit. Authentication is dangerously weak: a single, hard-coded, shared password means that discovering it on any one device compromises every device in the deployment, and provides no way to verify that a given message genuinely originates from a specific, individual device.

Worked Example

Problem 2. An attacker floods a smart building's IoT gateway with a very high volume of junk network traffic, preventing legitimate sensor readings from reaching the cloud. Name this threat type, and state which security requirement it violates.

Solution. This is a Denial-of-Service (DoS) attack, violating availability: the gateway remains technically uncompromised (no data is stolen or altered), but is prevented from performing its normal function for legitimate users.

Worked Example

Problem 3. A smart-agriculture deployment wants to guarantee that irrigation logs cannot be quietly altered after the fact, even by an employee with database access, without maintaining a full alternative copy of every reading on a separate system. Which technology from Section 6 addresses this requirement, and how, at a high level?

Solution. Recording a cryptographic hash of each irrigation log entry on a blockchain (Section 6.3) addresses this: altering the underlying stored log afterward would cause it to no longer match its previously recorded hash, making tampering detectable without needing to store a full duplicate copy of the data itself.

Worked Example

Problem 4. Explain, referencing Section 5, why an anomaly-detection model trained directly on raw, un-preprocessed multi-sensor data (with sensors reporting at different, irregular intervals) is likely to underperform a model trained on the same data after preprocessing and time-series structuring.

Solution. Without preprocessing, the model must contend with irregular sampling intervals, missing values, and potentially invalid readings as if they were genuine signal, making it harder to distinguish real anomalies from artefacts of inconsistent data collection; preprocessing and consistent time-series structuring remove or correct these artefacts first, so the model can focus on learning genuine patterns in the underlying physical process rather than patterns in the data-collection process itself.

8 Summary

IoT security rests on four core requirements – confidentiality, integrity, availability, and authenticity – realised through authentication, authorization, and encryption, and must be applied independently at the device, network, and cloud layers of the reference architecture from Module 1, since a defence at one layer leaves the others exposed. The threats this defends against – device attacks, eavesdropping, spoofing, denial-of-service, malware, and data leakage – are amplified in IoT specifically by device scale and frequently weak default security. Turning collected data into value requires a further pipeline of preprocessing, time-series structuring, visualization, and predictive analytics, each stage depending on the quality of the one before it. Finally, blockchain technology offers a decentralised alternative to the centralised trust model underlying conventional security, using smart contracts, tamper-evident data integrity, and self-sovereign identity management to remove single points of failure from device identity and data provenance – concepts that, together with everything from Modules 1–3, come together in the complete application domains of Unit V.

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5. Course notes, *Internet of Things (IoT) with Applications* – Module 4, IoT Security and Data Analytics.